

Operational semantics for Prolog with Cut in Rocq and its application to determinacy analysis

Jane Open Access   

Dummy University Computing Laboratory, [optional: Address], Country

My second affiliation, Country

Joan R. Public¹  

Department of Informatics, Dummy College, [optional: Address], Country

Abstract

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2012 ACM Subject Classification Replace ccsdesc macro with valid one

Keywords and phrases Dummy keyword

Digital Object Identifier 10.4230/LIPIcs.CVIT.2016.23

Funding *Jane Open Access:* (Optional) author-specific funding acknowledgements

Joan R. Public: [funding]

Acknowledgements I want to thank ...

1 Introduction

ELPI is a dialect of λ PROLOG (see [14, 15, 7, 12]) used as an extension language for the ROCQ prover (formerly the COQ proof assistant). ELPI has become an important infrastructure component: several projects and libraries depend on it [13, 3, 4, 19, 8, 9]. Examples include the Hierarchy-Builder library-structuring tool [5] and Derive [17, 18, 11], a program-and-proof synthesis framework with industrial applications at SkyLabs AI.

Starting with version 3, ELPI gained a static analysis for determinacy [10] to help users tame backtracking. ROCQ users are familiar with functional programming but not necessarily with logic programming and uncontrolled backtracking is a common source of inefficiency and makes debugging harder. The determinacy checkers identifies predicates that behave like functions, i.e., predicates that commit to their first solution and leave no *choice points* (places where backtracking could resume).

This paper reports our first steps towards a mechanization, in the ROCQ prover, of the determinacy analysis from [10]. We focus on the control operator *cut*, which is useful to restrict backtracking but makes the semantic depart from a pure logical reading.

We formalize two operational semantics for PROLOG with cut. The first is a stack-based semantics that closely models ELPI's implementation and is similar to the semantics mechanized by Pusch in ISABELLE/HOL [16] and to the model of Debray and Mishra [6, Sec. 4.3]. This stack-based semantics is a good starting point to study further optimizations used by standard PROLOG abstract machines [20, 1], but it makes reasoning about the scope of *cut* difficult. To address that limitation we introduce a tree-based semantics in which the branches pruned by *cut* are explicit and we prove the two semantics equivalent. Using the

¹ Optional footnote, e.g. to mark corresponding author



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42nd Conference on Very Important Topics (CVIT 2016).

Editors: John Q. Open and Joan R. Access; Article No. 23; pp. 23:1–23:13

Leibniz International Proceedings in Informatics



LIPICs Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl Publishing, Germany

```

Inductive P := IP of nat. Inductive D := ID of nat. Inductive V := IV of nat.

Inductive Tm :=
  | Tm_P of P
  | Tm_D of D
  | Tm_V of V
  | Tm_App of Tm & Tm.

Record Unif := {
  unify : Tm → Tm → Σ → option Σ;
  matching : Tm → Tm → Σ → option Σ;
}.

```

■ Figure 1 Tm types

■ Figure 2 Unif type

tree-based semantics we then show that if every rule of a predicate passes the determinacy analysis, the call to a deterministic predicate does not leave any choice points.

2 Common code: the language

Before going to the two semantics, we show the piece of data structure that are shared by the them. The smallest unit of code that we can use in the language is an atom. The atom inductive (see Type 1) is either a cut or a call. A call carries a term (see Figure 1). A term (Tm) is either a predicate, a datum, a variable or the binary application of two terms.

```

Inductive A := cut | call : Tm → A.
Record R := mkR { head : Tm; premises : list A }.
Record P := { rules : seq R; sig : sigT }.
Definition Σ := {fmap V → Tm}.
Definition bc : Unif → P →  $\mathcal{F}_v$  → Tm →
  Σ →  $\mathcal{F}_v$  * seq (Σ * seq A) :=

```

A rule (see Type 2) has a head of type term and a list of premises, that are a list of atoms. A program (see Type 3) is made by its rules and the signatures of predicates, **sigT** is a mapping from predicates to their signatures, signatures represent the definition of type from the simply typed lambda calculus, i.e. it is either a base type or an arrow for term application. We decorate arrows to know the mode (input or output) of the application argument.

A substitution (see Type 4) is a mapping from variables to terms. It is the output of a successful query. The empty substitution is noted ϵ .

The backchain function (bc, see Type 5) takes: a unificator U ; a program P ; a set fv of free variables ($\mathcal{F}_v$); a query q ; and the substitution σ . A backchain is performed when the current atom is a **call**. In this case, it start by recursively *dereferencing* the variables in σ , i.e. replacing all the variables in the query by their term mapping in σ . Then it takes the head of the term, which should be rigid, i.e. a predicate p , and look for the signature s of p in the program. Then, for each rule r in P , it unifies the head of r with q . Unification is possible thanks to the unificator U , whose type is shown in Figure 2. U explains how to unify terms up to standard unification (for output terms) or matching (for input terms). The function *unif* (resp. *matching*) unifies (resp. matches) the two terms under σ , it return an extension of σ if the two terms can be unified (resp. matched), and None otherwise. The result of a backchain is made by filtering the rules in P that can be used on the given query. In particular, for each rule filtered rules r , it returns a pair containing the extension of σ making the head of r and q unify and the list of premises of r .

2.1 The cut operator

ELPI is a dialect of λ PROLOG with the hard cut operator and, in the language we have shown before, *cut* is an atom. The semantics of the cut operator adopted in the ELPI language corresponds to the *hard cut* operator of standard SWI-PROLOG. This operator has two primary purposes. First, it eliminates all alternatives that are created either simultaneously with, or after, the introduction of the cut into the execution state.

The ELPI program we will use as an example in the following sections is shown in Figure 3.

```
f 1 2.    f 2 3.    r 2 4.    r 2 8.
g X X.    % r1
g X Z :- r X Z, !.    % r2
g X Z :- f X Y, f Y Z.    % r3
```

Figure 3 Small ELPI program example

To illustrate the high-level description of *cut*, consider the program shown in Figure 3 and the query $q = g\ 2\ Z$. All three rules for *g* can be used on the query q . They are tried according to their order of appearance in the program: rule r_1 is tried first, followed by r_2 , and r_3 .

The first rule has no premises and immediately returns the assignment $Z = 2$. However, the computation does not terminate at this point, since two additional unexplored alternatives remain, corresponding to the premises of rules r_2 and r_3 .

The premises of rule r_2 are $r\ 2\ Z, !$. At this stage, the role of the cut becomes apparent. If the premise $r\ 2\ Z$ succeeds, the cut commits to this choice and removes the premises of rule r_3 from the alternative list, as they were generated at the same point as the cut. Moreover, if the call $r\ 2\ Z$ itself produces multiple alternatives, only the first one is committed, while the remaining alternatives are discarded. This is because such alternatives have been created at a deeper depth in the search tree than the cut.

Concretely, the call $r\ 2\ Z$ yields two solutions, assigning Z the values 4 and 8, respectively. The second solution is eliminated by the cut, and only the first assignment is preserved.

3 Semantics intro

We propose two operational semantics for a logic program with cut. The two semantics are based on different syntaxes, the first syntax (called *tree*) exploits a tree-like structure and is ideal both to have a graphical view of its evolution while the tree is being interpreted and to prove lemmas over it. The second syntax, called *elpi*, is the ELPI's syntax and has the advantage of reducing the computational cost of cutting and backtracking alternatives by using shared pointers. We aim to prove the equivalence of the two semantics together with some interesting lemmas of the cut behavior.

4 Tree semantics

```
Inductive tree :=
  | KO | OK | TA of A
  | Or of option tree &  $\Sigma$  & tree
  | And of tree & seq A & tree.
```

se metti $r1 = g\ A\ B$. allora $B :- f\ A\ B$. allora $g\ e\ f$ sono funzioni, e puoi spiagare anche l'idea del detcheck qui

```

Fixpoint get_end s A :  $\Sigma$  * tree :=
  match A with
  | TA _ | KO | OK  $\Rightarrow$  (s, A)
  | Or None s1 B  $\Rightarrow$  get_end s1 B
  | Or (Some A) _ _  $\Rightarrow$  get_end s A
  | And A _ B  $\Rightarrow$ 
    let (s', pA) := get_end s A in
    if pA == OK then get_end s' B
    else (s', pA)
  end.

Definition get_subst s A := (get_end s A).1.
Definition path_end A := (get_end  $\epsilon$  A).2.
Definition success A := path_end A == OK.
Definition failed A := path_end A == KO.
Definition path_atom A := if path_end A is TA _ then true else

```

■ Figure 4 Definition of *get_end*

In the tree we distinguish 5 main cases: *KO*, *OK*, and are special meta-symbols representing, respectively, the basic failed and a successful trees. These symbols are considered meta because they are internal intermediate symbols used to give structure to the tree.

The *TA* constructor (acronym for tree-atom) is the constructor of atoms in the tree. They are either the cut operator or a call, in these cases, the interpreter should evaluate the atom by either cutting the subtrees in the scope of the cut or backchaining the call in the program.

The two recursive cases of a tree are the *Or* and *And* nodes. The *Or* node $A \vee B_\sigma$ denotes a disjunction between two trees *A* and *B*. The first branch is optional, if absent it represents a dead tree, i.e. a tree that has been entirely explored. The second branch is annotated with a suspended substitution σ so that, upon backtracking to *B*, σ is used as the initial substitution for the execution of *B*.

The *And* node $A \wedge_{B_0} B$ represents a conjunction of two trees *A* and *B*. We call B_0 the reset point for *B*; it is used to restore the state of *B* to its initial form if a backtracking operation occurs on *A*. Intuitively, let *t2l* be the function flattening a tree in a list of sequents disjunction, in PROLOG-like syntax the tree $A \wedge_{B_0} B$ becomes $(A_1, t2l B); (A_2, B_0); \dots; (A_n, B_0)$ where $t2l(A) = A_1, \dots, A_n$.

A graphical representation of a tree is shown in Figure 5a. To make the graph more compact, the *And* and *Or* nodes are n-ary rather than binary, and associate to the right. The *KO* node acts as the neutral elements in the *Or* list, while *OK* is the neutral element of the *And* list.

The interpretation of a tree is performed by two main routines: *step* and *next_alt* that traverse the tree depth-first, left-to-right. Then, then *run* inductive makes the transitive closure of step *step* and *next_alt*: it iterates the calls to its auxiliary functions. In Types 7–9 we give the types contrats of these symbols where *fvS* is a set of variable names.

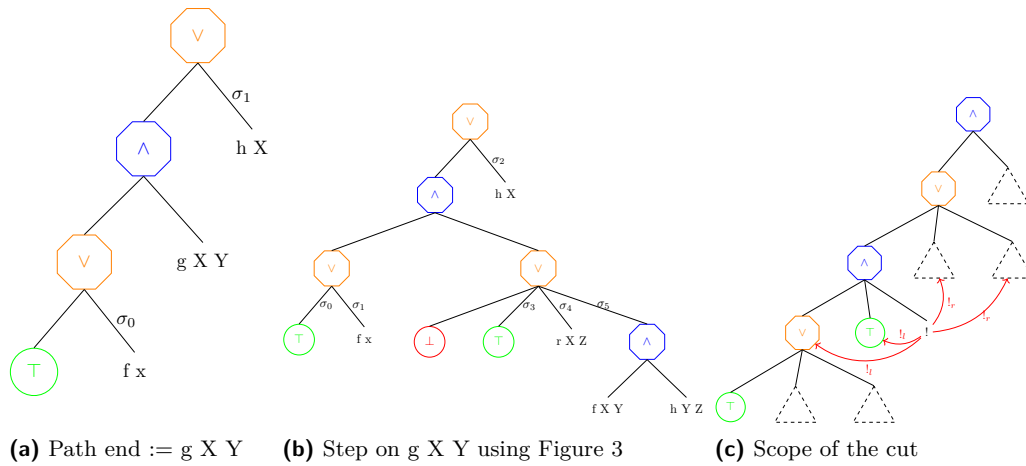
Inductive tag := Expanded | CutBrothers | Failed | Success. (6)

Definition step : $\mathbb{P} \rightarrow \mathcal{F} \rightarrow \Sigma \rightarrow \text{tree} \rightarrow (\mathcal{F} * \text{tag} * \text{tree}) :=$ (7)

Definition next_alt : $\mathbb{B} \rightarrow \text{tree} \rightarrow \text{option tree} :=$ (8)

Inductive run (u:Unif) (p : $\mathbb{P}$): $\mathcal{F} \rightarrow \Sigma \rightarrow \text{tree} \rightarrow \Sigma \rightarrow \text{option tree} \rightarrow \text{Prop} :=$ (9)

The tree interpreter, as in prolog, explores the tree in DFS strategy, to discover the substitution and the leaf of the tree that should be interpreted. The *get_end* routine, shown in Figure 4, accomplishes to this task. The *get_end* returns its inputs if the tree is a leaf. Otherwise, if the tree is a disjunction, the path continues on the left subtree, if it exists, otherwise it recursively retrieves the wanted piece of information in the rhs using the substitution stored in the *Or* branch: the current substitution when we cross the rhs of a *Or* is the one store in the *Or* node itself. In the case of a conjunction, if the to-be-explored leaf in



■ **Figure 5** Some tree representations

139 the lhs is *OK*, then we look for the *get_end* in the rhs, otherwise we return the result of the
 140 lhs.

141 We derive the following two functions from *get_end*:

142 **Definition** *get_subst* s A := (*get_end* s A).1. (1)

143 **Definition** *path_end* A := (*get_end* A).2. (2)

144 In Figure 5a the *path_end* of the tree is g X Y.

145 Below we define three special kinds of trees depending on their *path_end*.

146 **Definition** *success* A := *path_end* A == OK. (3)

147 **Definition** *failed* A := *path_end* A == KO. (4)

148 **Definition** *path_atom* A := if *path_end* A is TA _ then true else false. (5)

149 The latter definition identifies path ending in an atom.

150 4.1 The *step* procedure

151 The *step* procedure takes as input a program, a set of free variables (fv), a substitution, and
 152 a tree, and returns an updated set of free variables, a *step_tag*, and an updated tree.

153 Free variables are those variables that appear in a tree; they are used and updated when
 154 a backchaining operation takes place.

155 The *step_tag* (see Type 6) indicates the kind of an internal tree step: **CutBrothers** denotes
 156 the interpretation of a superficial cut, i.e., a cut whose parent nodes are all *And*-nodes.
 157 **Expanded** denotes the interpretation of non-superficial cuts or predicate calls. **Failure** and
 158 **Success** are returned for, respectively, *failed* and *success* trees.

159 The step procedure is intended to interpretate atoms, that is, it transforms the tree iff its
 160 *path_end* is an atom, otherwise, it returns the identity.

161 **Lemma** *succ_step_iff* u p fv s A: *success* A $\leftrightarrow$ *step* u p fv s A = (fv, *Success*, A). (1)

162 **Lemma** *fail_step_iff* u p fv s A: *failed* A $\leftrightarrow$ *step* u p fv s A = (fv, *Failed*, A). (2)

163 *Call step* The interpretation of a call *c* starts by calling the *bc* function on *c*. The output
 164 list *l* is taken to represent build the new subtree. If *l* is empty then *KO* tree is returned,
 165 otherwise the subtree is a right-skewed tree made of *n* inner *Or* nodes, where *n* is the length
 166 of *l*. The root has *KO* as left child. The lhs of the other nodes is a right-skewed tree of *And*
 167 nodes. The *And* nodes are again a right-skewed tree containing premises of the selected rule.

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168 A step in the tree of Figure 5a makes a backchain operation over the query $g \ X \ Y$ and, in
 169 the program defined in Figure 3, the new tree would be the one in Figure 5b. We have put a
 170 red border around the new generated subtree. It is a disjunction of four subtrees: the first
 171 node is the *KO* node (by default), the second is *OK*, since r_1 has no premises, the third and
 172 the fourth contains the premises of respectively r_2 and r_3 .

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Cut step The cut case is delicate, since interpreting a cut in a tree has three main impacts:
 at first the cut is replaced by the *OK* node, then some special subtrees, in the scope of the
Cut, are cut away: in particular we need to soft-kill the left-siblings of the *Cut* and hard-kill
 the right-uncles of the the *Cut*.

177 ► **Definition 1** (Left-siblings (resp. right-sibling)). *Given a node A , the left-siblings (resp.*
 178 *right-sibling) of A are the list of subtrees sharing the same parent of A and that appear on*
 179 *its left (resp. right).*

180 ► **Definition 2** (Right-uncles). *Given a node A , the right-uncles of A are the list of right-sibling*
 181 *of the father of A .*

182 ► **Definition 3** (Hard-kill, $!_r$). *Given a tree t , hard-kill replaces the given subtree with the*
 183 *KO node*

184 ► **Definition 4** (Soft-kill, $!_l$). *Given a successfull tree t , soft-kill replaces with KO all subtrees*
 185 *that are not part of the path in t leading to the OK node.*

186 An example of the impact of the cut is show in Figure 5c, the dashed triangles represent
 187 generic trees. The step routine interprets the cut since it is the node in its path-end: we pass
 188 through a and all trees on the left of the cut are successful. In the example we have 4
 189 arrow tagged with the $!_l$ or $!_r$ symbols. The $!_l$ arrows go left and soft-kill the pointed subtree,
 190 it keeps *OK* nodes since they are part of the tree leading to the cut, and replaces the other
 191 subtrees with *KO*. The $!_r$ procedure replaces the nodes pointed by the arrows with *KO*.

192 4.2 The *next_alt* procedure

193 It is evident that the *step* alone is not sufficient to reproduce entirely the behavior of the
 194 full expected prolog interpreter. In particular, we need to bracktrack on failures. Moreover,
 195 in case of success, we should return a tree cleaned of its success, this is essential to, non
 196 deterministically, find the next solution of a given query. By Lemmas 1 and 2, we know that
 197 *step* returns the identity on successful and failed trees. In order to continue the computation
 198 on these particular trees, we need the *next_alt* procedure aiming to expecially work with
 199 failed and successful trees: and its implementation in Figure 6.

200 The *next_alt* procedure takes a boolean and a tree, clean it from failures or success and
 201 returns a new tree if this tree still contains a non explored path. The idea behind *next_alt* is
 202 to clean recursively every subtree in DFS order if its *path_end* is a failure. Moreover, if the
 203 boolean passed to *next_alt* is true, then it erases the first successful path in the tree.

204 The base cases of *next_alt* are immidiate. The *Or* case is rathere intuitive: if the lhs
 205 of the *Or* does not exist we look for the *next_alt* in the rhs. Otherwise, we look for the
 206 *next_alt* in the lhs, if this *next_alt* does not exists, we look for the *next_alt* in the rhs.

207 We want to spend few words about the *And* case, since the reset point B_0 for B plays an
 208 important role. The *next_alt* in an *And* tree should consider two cases: if the lhs succeeds,
 209 then the *next_alt* should be retrived in the rhs. If this alternative does not exists it means
 210 that the rhs has entirely been explored. We need to erase the success in the lhs and try to
 211 find if a non-explored alternative exists. If so, we return a new tree with the new lhs and the

```

Definition next_alt :  $\mathbb{B} \rightarrow \text{tree} \rightarrow \text{option tree} :=
  \text{fix next\_alt } b \ A :=
  \text{match } A \text{ with}
  | KO \Rightarrow \text{None}
  | OK \Rightarrow \text{if } b \text{ then None else Some OK}
  | TA _ \Rightarrow \text{Some } A
  | And A BO B \Rightarrow
    \text{let build\_BO } A := \text{And } A \text{ BO } (\text{big\_and } BO) \text{ in}
    \text{if success } A \text{ then}
      \text{match next\_alt } b \ B \text{ with}
      | None \Rightarrow \text{omap build\_BO } (\text{next\_alt true } A)
      | Some B' \Rightarrow \text{Some } (\text{And } A \text{ BO } B')
    \text{end}
    \text{else if failed } A \text{ then omap build\_BO } (\text{next\_alt false } A)
    \text{else Some } (\text{And } A \text{ BO } B)
  | Or None sB B \Rightarrow \text{omap } (\text{fun } x \Rightarrow \text{Or None sB } x) (\text{next\_alt } b \ B)
  | Or (Some A) sB B \Rightarrow
    \text{match next\_alt } b \ A \text{ with}
    | None \Rightarrow \text{omap } (\text{fun } x \Rightarrow \text{Or None sB } x) (\text{next\_alt false } B)
    | Some A' \Rightarrow \text{Some } (\text{Or } (\text{Some } A') \text{ sB } B)
    \text{end}
  \text{end.}$ 
```

■ **Figure 6** *next_alt* implementation

rhs is built from the reset point. *big_and* is a trivial function build a right-skewed tree of and nodes where the leaves are the atoms written in the reset point. We need to reuse the reset point since, the step procedure in *And* trees evaluates the rhs of a *And* tree if the lhs succeeds. This evaluation is dependent on the substitution in the lhs tree. Therefore, if we need to backtrack in the lhs, we need to reset the rhs.

Some interesting property of *next_alt* are shown below and allow to see how *next_alt* complements *step*.

Lemma *path_atom_next_alt_id* $b \ A$: $\text{path_atom } A \rightarrow \text{next_alt } b \ A = \text{Some } A$. (3)

Lemma *next_alt_failedF* $b \ A \ A'$: $\text{next_alt } b \ A = \text{Some } A' \rightarrow \text{failed } A' = \text{false}$. (4)

For example, in Figure 5b the step procedure has created a failed tree: its path-end ends in *KO*. The expected behavior of *next_alt* is to take this *KO* node and make it a This allows *step* to continue the exploration of the tree. In particular, the path-end of this new tree ends in *OK*. The step leaves the tree unchanged producing the new substitution. This solution however is not unique, we should be able to backtrack on this successful tree. To do so we can call *next_alt* and it will deadify the *OK* node allowing *step* to proceed on $r \ X \ Z$.

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4.3 The *run* inductive

The inductive procedure *run* is modeled as a function: it takes as input a program, a set of free variables, an initial substitution σ_0 , and a tree t_0 , and returns a substitution σ_1 together with an optional updated tree t_1 . The substitution σ_1 represents the most-general unificator that makes the execution of the tree t_0 succeed starting from the initial substitution σ_0 , σ_1 is an extension of σ_0 . The tree t_1 is the updated tree containing the alternatives that have not yet been explored. If the tree contains no solution, then *None* is returned.

The procedure *run* is based on three main derivation rules, shown in Figure 7. If the *path_end* of the tree *t* is a success, the input substitution is returned and the input tree is cleaned of its successful path. If the *path_end* of the tree is an atom, then *step* is invoked to evaluate this atom, and *run* is recursively called on the new tree. Finally, if the *path_end* of the tree is a failure, *next_alt* is called to clear the failed path; if the resulting cleaned tree exists, *run* is recursively called on it.

$$\begin{array}{c}
\frac{\text{success } A \quad \text{get_subst } s_1 \ A = s_2 \quad (\text{next_alt true } A) = B}{\text{run fv } s_1 \ A \ s_2 \ B} \text{ RUN_DONE} \\
\\
\frac{\text{path_atom } A \quad \text{step u p fv0 } s_1 \ A = (\text{fv1}, \text{st}, B) \quad \text{run fv1 } s_1 \ B \ s_2 \ r}{\text{run fv0 } s_1 \ A \ s_2 \ r} \text{ RUN_STEP} \\
\\
\frac{\text{failed } A \quad \text{next_alt false } A = \text{Some } B \quad \text{run fv0 } s_1 \ B \ s_2 \ r}{\text{run fv0 } s_1 \ A \ s_2 \ r} \text{ RUN_FAIL}
\end{array}$$

■ **Figure 7** Rule system for *run*

239

5 Elpi semantics

240

We now want to introduce the elpi semantics. The interpreter we show reflects the interpreter of the ELPI language and is an operational semantics close to the one picked by Pusch in [16], in turn closely related to the one given by Debray and Mishra in [6, Section 4.3]. Pusch mechanized the semantics in Isabelle/HOL together with some optimizations that are present in the Warren Abstract Machine [20, 1].

The inductive representing a state of the ELPI language is shown below.

```

Inductive alts := no_alt | more_alt of (Σ * goals) & alts
with goals := no_goals | more_goals of (A * alts) & goals .

```

An elpi state is an enhanced two-dimension list. The outermost list represents the list of alternatives in disjunction accompagnate with the substitution that should be used to for their interpretation. The innermost list is a list of atom, representing a list of goals in conjunctions. These goals are decorated with a pointer to an elpi state, and are used to keep trace of the alternatives that should be kept when a cut is interpreted. We call these, special, alternatives the cut-to alternatives.

The idea of the ELPI interpreter is to receive a list of alternatives. The first alternative consists of a list of goals. Four cases must be taken into account; they are shown in Figure 8. In order to simplify goal retrieval, we split the head of the alternatives from the tail, so that it can be immediately matched in the inductive definition. Note that an empty list of alternatives represents, by definition, a failing elpi state. If the goal list is empty (STOPE), then we have, by definition, a success, and the input solution together with the list of alternatives is returned. If the goal list starts with a cut (CUTE), then the current alternatives are erased in favour of the cut alternatives, and a recursive call is made on the remaining goal list.

Finally, we must consider the case in which the goal list starts with a call. The call can either fail (FAILE) or succeed (CALLE). We distinguish the two cases by looking if the backchaining operation returns zero or more rules. We have wrapped this task in the *stepE*

Definition $\text{stepE } \text{fv } t \text{ s } a \text{ gl} :=$
 $\text{let } (\text{fv}', \text{rs}) := \text{bc_up } \text{fv } t \text{ s } \text{ in}$
 $\text{let } \text{rs_ca} := \text{save_alts } a \text{ gl } (\text{r2a } \text{rs}) \text{ in}$
 $(\text{fv}', \text{rs_ca}).$

Inductive $\text{nur} : \mathcal{F}_v \rightarrow \Sigma \rightarrow \text{goals} \rightarrow \text{alts} \rightarrow \Sigma \rightarrow \text{alts} \rightarrow \text{Prop} :=$ (10)

$$\frac{}{\text{nur } \text{fv } s \text{ } [::] \text{ } a \text{ } s \text{ } a} \text{STOPE}$$

$$\frac{\text{nur } \text{fv } s \text{ gl } \text{ca } s_1 \text{ r}}{\text{nur } \text{fv } s \text{ } [:: (\text{cut}, \text{ca}) \ \& \ \text{gl}] \text{ } a \text{ } s_1 \text{ r}} \text{CUTE}$$

$$\frac{\text{stepE } \text{fv } t \text{ s } \text{al } \text{gl} = (\text{fv}', [:: \text{b} \ \& \ \text{bs}]) \quad \text{nur } \text{fv}' \text{ b.1 } \text{b.2 } (\text{bs}++\text{al}) \text{ } s_1 \text{ r}}{\text{nur } \text{fv } s \text{ } [:: (\text{call } t, \text{ca}) \ \& \ \text{gl}] \text{ } \text{al } s_1 \text{ r}} \text{CALLE}$$

$$\frac{\text{stepE } \text{fv } t \text{ s } \text{al } \text{gl} = (\text{fv}', [::]) \quad \text{nur } \text{fv}' \text{ } s_1 \text{ } a \text{ } \text{al } s_2 \text{ r}}{\text{nur } \text{fv } s \text{ } [:: (\text{call } t, \text{ca}) \ \& \ \text{gl}] \text{ } [:: (s_1, a) \ \& \ \text{al}] \text{ } s_2 \text{ r}} \text{FAILE}$$

■ **Figure 8** Rule system for *nur*

265 procedure, which also updates the goal and cut-alternative list. The fail case, is relatively
 266 easy: the first goal does not succeed, we need to take the head of the alternatives, and make
 267 it the new list of goals to be explored.

268 The case in which backchaining produces a non empty list, the *save_alts* routine is in
 269 cahрге of: taking the list of premises and add to each atom the the list of alternatives *a* as
 270 their new cut-alternatives, then it append the list of goals *gl* to each of these new lists.

271 6 Semantic equivalence

272 The equivalence between the two semantics is possible under two conditions: we need to
 273 work with “valid tree”, i.e. all of those trees that can be generated from a call. Secondly,
 274 we need to translate trees into elpi states. In the next to subsection we propose the two
 275 functions *valid_state* and *t2l*.

276 6.1 Valid trees

277 The inductive tree allows one to generate a large number of trees, some of which are not
 278 valid, in the sense that they cannot be produced starting from a given query. The class of
 279 valid trees is characterized by the function shown in Figure 9a.

280 While all base cases of tree are considered valid, we need to analyze carefully the cases
 281 for the *Or* and *And* constructors.

282 For the *Or* constructor, we distinguish two cases depending on whether the left-hand
 283 side (lhs) exists. If it does not exist, then the right-hand side (rhs) must be a valid tree.
 284 Otherwise, the lhs must itself be a valid tree, and the rhs is either the *KO* tree, since it may
 285 have been removed by the evaluation of a superficial cut in the lhs, or it has not yet been
 286 explored. In the latter case, it is a *base_or* tree, namely the right-skewed tree formed by a
 287 disjunction of conjunctions that is generated after a successful backchain to a call.

```

Fixpoint t2l A s0 bt :=
  match A with
  | OK           ⇒ [:: (s0, [::])]
  | KO           ⇒ [::]
  | TA a         ⇒ [:: (s0, [:: (a, [::]) ])]
  | Or None s1 B ⇒ add_ca_deep bt (t2l B s1 [::])
  | Or (Some A) s1 B ⇒
    let lB := t2l B s1 [::] in
    let lA := t2l A s0 lB in
    add_ca_deep bt (lA ++ lB)
  | And A B0 B ⇒
    let lA := t2l A s0 bt in
    let lB0 := a2g B0 in
    let lA := add_deep bt lB0 lA in
    if lA is [:: (s0, gs) & al] then
      let al := map (catr lB0) al in
      let lB := t2l B s0 (al ++ bt) in
      map (catl gs) lB ++ al
    else [::]
  end.
end.

Fixpoint valid_tree A :=
  match A with
  | TA _ | OK | KO ⇒ true
  | Or None _ B ⇒ valid_tree B
  | Or (Some A) _ B ⇒ valid_tree A &&
    ((B == KO) || B.base_or B)
  | And A B0 B ⇒ valid_tree A &&
    if success A then valid_tree B
    else B == big_and B0
  end.

```

(a) Valid tree

(b) Tree to list

■ **Figure 9** Valid tree and Tree to list

For the *And* constructor, the lhs is required to be a valid tree. The shape of the rhs depends on whether the lhs is a success. If the lhs is not successful, then the rhs has never been explored: the procedures *step* and *next_alt* modify the rhs only when the lhs succeeds. In this case, the lhs must be the right-skewed tree containing conjunctions of premises atoms, the reset point B_0 ensure what shape the rhs should have. If the lhs is a success tree, then the rhs can be modified by *step* and *next_alt*, therefore it must be a valid tree.

6.2 From trees to lists

The translation of a tree to a list is shown in Figure 9b. It takes the tree to be translated, a substitution (called s), a list of alternatives, called bt . The substitution s tells what is the substitution of the alternative we are building and is updated when going to the rhs of the *Or* constructor. The bt list represents the future alternatives list that have the same depth of the current tree root. They are useful to know if they should be added to the current goal as cut alternatives.

An *OK* node represent a success in a tree, then the translation of this tree returns a singleton list with the couple $(s, [::])$: there is one alternative with 0 goals, i.e. a success in elpi by *StoPE*. The *KO* node represent a failure, and, therefore 0 alternatives. An atom is translated into a singleton with the atom as first goal and the empty cut-alternative list: we are not adding bt since they are the alternatives for the right-sibling, not the right-uncles, this means that if we have a is a cut, then it erases the alternatives bt .

In a disjunction, we are scendendo di un livello the distance from our backtracking list. This means that bt becomes the list of right-uncles of the two branches of the *Or* constructor. The compilation of the rhs is done independently of if the lhs exists: we transform the rhs into an elpi state and passing the empty list of alternatives, since the rhs as no right-siblings. Then, thanks to *add_ca_deep*, we recursively add bt to every cut-alternative appearing in

the translated goals. If the lhs exists, we translate it and pass the translation of the rhs as the list of right-siblings. Then we prepend the translated lhs to the translated rhs and add bt with `add_ca_deep`.

We compile the *And* case by translating its lhs to the list lA and reset point to the list $lB0$. Then, thanks to `add_deep`, we recursively append $lB0$ to the alternatives born in lA , i.e. we leave unchanged the pointers to bt , if any, in lA . We finally need to treat cases whenever the list lA is empty or not. In the former case the empty list is returned, otherwise, we have a list $[(s_0, gs) \& al]$. By the semantics of the *And* in the *run* interpreter, the rhs of the *And* represent the sequent of goals to be executed after the first alternative in the lhs, it corresponds to the list of goals gs . The reset point $B0$ is to be appended to the next alternatives in the lhs, that is the queue of lA , that is al .

alternative in tree? rephrase

We note therefore that the compilation of lB is done by passing the alternatives $al + +bt$ since the alternatives in al are to be...

6.3 Equivalence theorems

We have now all the ingredients that ...

```

Lemma tree_to_elpi:  $\forall$  u p  $s_0$  t  $s_2$  t',
  let fv := vars_tree t `|` vars_sigma  $s_0$  in
  valid_tree t  $\rightarrow$ 
    run u p fv  $s_0$  t  $s_2$  t'  $\rightarrow$ 
       $\exists$   $s_1$  g a,
        let: na := (t2l (odflt K0 t')  $s_0$  [::]) in
          t2l t  $s_0$  [::] = ( $s_1$ , g) :: a /\
            nur u p fv  $s_1$  g a  $s_2$  na.

```

```

Lemma elpi_to_tree:  $\forall$  u p fv  $s_1$  g  $s_2$  a na,
  nur u p fv  $s_1$  g a  $s_2$  na  $\rightarrow$ 
     $\forall$   $s_0$  t, valid_tree t  $\rightarrow$  t2l t  $s_0$  [::] = ( $s_1$ , g) :: a  $\rightarrow$ 
       $\exists$  t', run u p fv  $s_0$  t  $s_2$  t' /\ t2l (odflt K0 t')  $s_0$  [::] = na.

```

The proof of Lemma 6 is based on the idea explained in [2, Section 3.3]. An ideal statement for this lemma would be: given a function `l2t` transforming an elpi state to a tree, we would have have that the the execution of an elpi state e is the same as executing *run* on the tree resulting from `l2t(e)`. However, it is difficult to retrieve the structure of an elpi state and create a tree from it. This is because, in an elpi state, we have no clear information about the scope of an atom inside the list and, therefore, no evident clue about where this atom should be place in the tree.

Our theorem states that, starting from a valid tree t which translates to a list of alternatives $(\sigma_1, g) :: a$. If we run in elpi the list of alternatives, then the execution of the tree t returns the same result as the execution in elpi. The proof is performed by induction on the derivations of the elpi execution. We have 4 derivations.

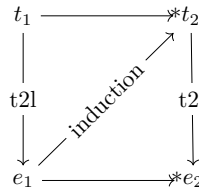
We have 4 case to analyse:

7 Case study: determinacy analysis

we mechanize the first order part of xxx.

snippet det, main thm, invariant det tree (valid tree prev section?)

proof inducion on exec, step/next alt preserving invariant proved by induction on the tree.



■ **Figure 10** Induction scheme for Lemma 6

346 with list semantics cut and next alt requires to express a link between the ca or next alts
 347 and the current goal, which is non trivial without an intermediate data strature like the tree

348 8 Related work

349 prolog semantics, King lost
 350 yves for the proof technique

351 9 Conclusion

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